



Manthan



AUTONOMOUS BILLING-DISPUTE INVESTIGATION • LIVE ON GOOGLE CLOUD

Track 3 • Google for Startups AI Agents Challenge

Refactor for Google Cloud Marketplace & Gemini Enterprise

live app manthan-ui-dzv6bwbpbpa-uc.a.run.app
[card.json](#)

agent card [manthan-advisor-dzv6bwbpbpa-uc.a.run.app/.well-known/agent-](https://manthan-advisor-dzv6bwbpbpa-uc.a.run.app/.well-known/agent-code)
code github.com/Miny-Labs/manthan

WHY IT EXISTS

Scattered truth. Drowning in tokens.

A billing dispute — chargeback, refund demand, failed payment, fraud warning — lands with a deadline. The evidence is spread across the merchant's stack: the charge in Stripe, the account in the CRM, the complaint in support, the outage in observability, the refund formula in the policy docs.

Someone must read all of it, apply documented policy, and compute what is actually owed — before the clock runs out.

9

systems hold
the evidence

~5 h

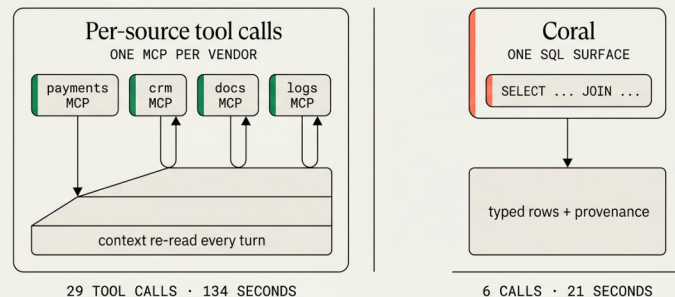
analyst hours
per dispute

~43×

naive-loop
token cost

The bill is an architecture decision

SAME TASK · SAME MODEL · DIFFERENT PLUMBING



82-TASK BENCHMARK: 64% FEWER TOKENS · 70% CHEAPER · 55% FASTER · 31% MORE ACCURATE

The bill is an architecture decision: one SQL surface over nine systems beats per-vendor tool chains — 64% fewer tokens, 70% cheaper on Coral's published 82-task benchmark.

Three agents are the system

1 • Trigger

Stripe webhook (5 event types) hits the triage agent — or any agent calls `investigate_dispute` over A2A.

2 • Investigate

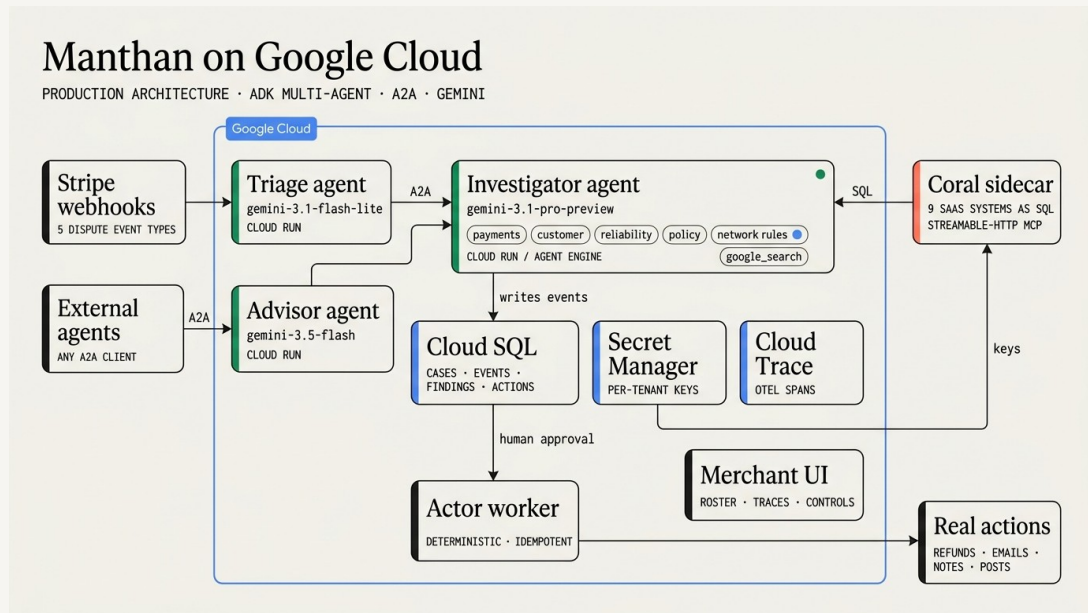
Triage dispatches the investigator over A2A. A coordinator fans out five specialists in parallel; every claim is cited to its source record.

3 • Decide

Fight, refund, accept, or escalate — with the math shown. The agent only ever proposes.

4 • Approve & act

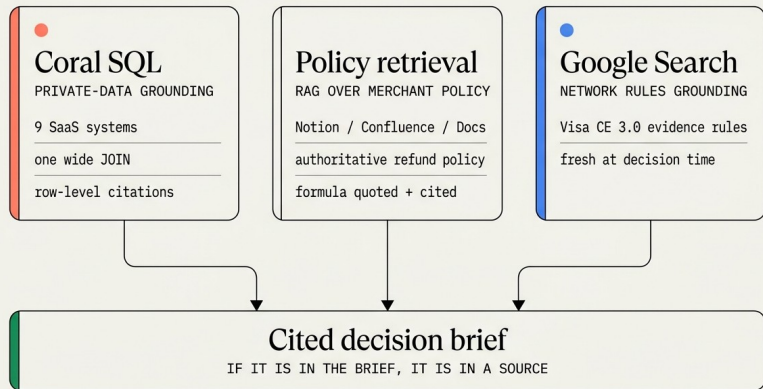
Policy gates route to a human tier (auto <\$50 • one-click • two-person). A deterministic actor executes with idempotency keys — the LLM never holds write credentials.



Grounded in the systems of record

Three grounding surfaces

PRIVATE DATA · POLICY DOCS RAG · LIVE WEB



Private data — Coral SQL

Nine live systems as Postgres-compatible schemas behind one MCP server. Row-level provenance: every citation chip deep-links to the record. Read-only by design; the model sees rows, never keys.

RAG — merchant policy

The policy analyst retrieves the merchant's own SOPs — Notion, Confluence, or Docs, discovered from the catalog — and quotes the refund formula verbatim.

Google Search — network rules

Card-network evidence requirements (Visa CE 3.0 per reason code) via ADK's built-in `google_search` — too volatile to hardcode, fetched fresh for fight briefs.

Coral vs per-vendor MCPs (82 tasks): 31% more accurate · 64% fewer tokens · 70% cheaper · 55% faster

Built on the agent platform, end to end

Google ADK 2.x

Coordinator + five specialists as AgentTools; governance rules as before_model / before_tool callbacks

Gemini — three tiers

3.1-pro coordinator · 3.5-flash specialists + advisor · 3.1-flash-lite triage.
Every reasoning call is Gemini

A2A protocol

Three Agent Cards, JSON-RPC /a2a. Internal dispatch and the external surface: 12 skills for other agents

Cloud Run × 6

Gateway, three agent services, worker, UI — scale-to-zero budget profile (~\$13/mo)

Cloud SQL + Secret Manager

Postgres projections · per-tenant secrets with per-secret IAM — agents read only what they are wired to

Agent Identity

One service account per agent; identity block (agent id, SA, model, fingerprint) published on every card

Cloud Trace

OpenTelemetry end to end — spans for every model call, tool call, and specialist; live span tree in the UI

Agent Engine path

The investigator runs the same ADK code on Vertex AI Agent Engine (documented); Cloud Run is the live default

PROVEN IN PRODUCTION

A real dispute, end to end, unassisted

200 s

webhook → verdict

27

tool calls · 8 systems

5

cited findings

Thin evidence → escalate @ 0.80

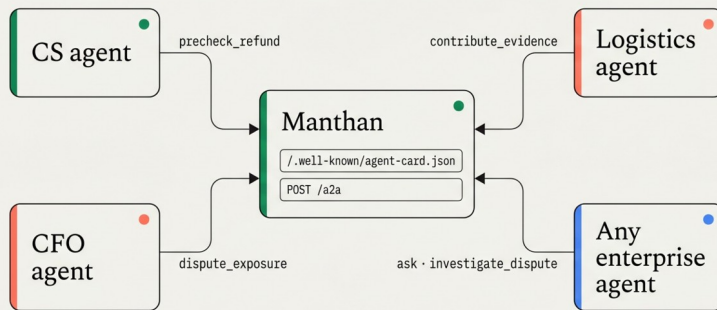
Synthetic dispute, no customer trail. The agent recorded “livemode: false — test transaction” as a finding instead of fabricating, and routed to a human.

Rich evidence → refund exactly \$560

Seeded \$8,400 case: the pacer rejected the first conclusion for missing math; the agent derived $\$8,400 \div 30 \times 2$ degraded days, cited it, then concluded @ 0.95.

Any agent can work with Manthan

A2A · AGENT CARD + JSON-RPC · 12 SKILLS



EVERY CASE ARTIFACT IS READABLE: case · brief · findings · actions · audit

The agent knows what it doesn't know.

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